PRINCIPAL COMPONENT ANALYSIS (PCA)

Dimensionality Reduction in Business

Any process in business is multidimensional:

for instance:

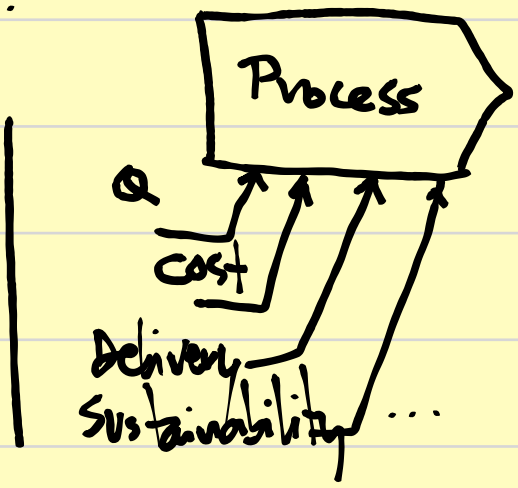
KPI

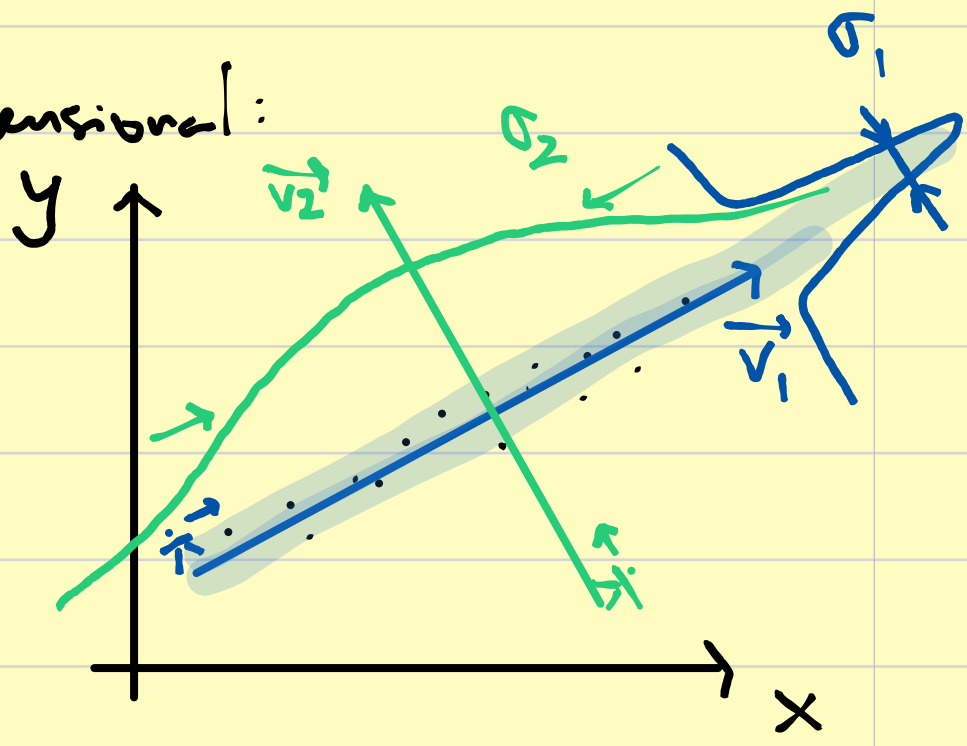
ppm

€

:

CO2 ...





The direction that shows the minimal σ is called PRINCIPAL COMPONENT #1. PC_1

Perpendicular to PC_1, we have PC_2 (second PC). ($\sigma_2 \gg \sigma_1$)

We can describe the variability of the process with only one PC and still understand much of it.

DEFINITION PC. The PCs are the EIGENVECTORS of the COVARIANCE MATRIX.

The covariance Matrix of a system with "n" variables

$$cov[x_1, \dots, x_n] = \begin{bmatrix} VAR[x_1] & \cdots & cov[x_1, x_n] \\ \vdots & \ddots & \vdots \\ cov[x_1, x_n] & & VAR[x_n] \end{bmatrix} \quad i, j = 1, \dots, n$$

$$VAR[x_i] = \frac{\sum (x_i - \bar{x})^2}{n-1} \qquad cov[x_i, x_j] = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

Example: A company has a Mgmt System with two KPIs (Lead Time & Cost per Unit). Please calculate the Covariance matrix given the time series:

$LT[days]: \{17, 19, 21, 27\} \equiv X$

$COST[€/Unit]: \{7, 11, 12, 15\} \equiv Y$

$$\bar{X} = \frac{17+19+21+27}{4} = 21 \; ; \; \bar{Y} = \frac{7+11+12+15}{4} = 11{,}25$$

$$VAR[X] = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{(17-21)^2 + (19-21)^2 + (21-21)^2 + (27-21)^2}{4-1} = 18{,}7$$

$$VAR[Y] = \frac{\sum (y_i - \bar{y})^2}{n-1} = \frac{(7-11{,}25)^2 + (11-11{,}25)^2 + (12-11{,}25)^2 + (15-11{,}25)^2}{4-1} = 10{,}9$$

$$cov[X,Y] = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1} = \frac{(17-21)(7-11{,}25) + (19-21)(11-11{,}25) + (21-21)(12-11{,}25) + (27-21)(15-11{,}25)}{4-1} = 13{,}3$$

$$cov_{MAT}[X,Y] = \begin{bmatrix} 18{,}7 & 13{,}3 \\ 13{,}3 & 10{,}9 \end{bmatrix} \equiv A$$

Exams are parametrized:

Matrikelnummer $\underline{2}\;\underline{3}\;\underline{7}\;\underline{6}\;\underline{3}\;\underline{9}$

$a \; b \; c \; d \; e \; f$

$LT\{7+2a, 13+3b, \dots\}$

$\downarrow$

$LT\{7+4, 22, \dots\}$

· Now we calculate the Principal Components of the management system [Eigenvectors of the Cov. MAT.].

Eigenvectors & Eigenvalues $[\vec{v}, \lambda]$

$$A\vec{v} = \lambda\vec{v} \equiv \det[A - \lambda I] = 0 \rightarrow \lambda \rightarrow \vec{v}$$

$$\det\left[\begin{bmatrix} 18'7 & 13'3 \\ 13'3 & 10'9 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right] = 0 \rightarrow \det\begin{bmatrix} 18'7-\lambda & 13'3 \\ 13'3 & 10'9-\lambda \end{bmatrix} = 0 \xrightarrow{(\cdot)}$$

(·)

$$\det\begin{bmatrix} a & c \\ b & d \end{bmatrix} = a.d - b.c$$

$$\rightarrow (18-\lambda)(10'9-\lambda) - 13'3^2 = 0 \rightarrow$$

$$\rightarrow \lambda^2 - 29'6\lambda + 26'2 = 0 \xrightarrow{(\cdot\cdot)}$$

$$ax^2+bx+c=0 \rightarrow x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$\lambda = \frac{29'6 \pm \sqrt{29'6^2 - 4.26'2}}{2} \nearrow \lambda_1 = 28'7$$

$$\searrow \lambda_2 = 0'9$$

(··)

The relative magnitude of the Eigenvalues represents the amount of variability explained by each Eigenvector.

How much variability does $\vec{v_1}$ explain? $\%VAR\vec{v_1} = \frac{28'7}{28'7+0'9} = 96'9\%$

$\vec{v_2}$ $\%VAR\vec{v_2} = \frac{0'9}{28'7+0'9} = 3'1\%$

We can already say that with only the first PC_1, we can explain 96'9 % of the variability of the process.

$\boxed{\lambda_1 = 28'7} \longrightarrow A\vec{v}_1 = \lambda_1 \vec{v}_1 \longrightarrow \begin{bmatrix} 18'7 & 13'3 \\ 13'3 & 10'9 \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = 28'7 \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} \longrightarrow$

$18'7 \cdot v_{11} + 13'3 \cdot v_{12} = 28'7\, v_{11}$

$13'3 \cdot v_{11} + 10'9 \cdot v_{12} = 28'7\, v_{12}$ $\Rightarrow$ $v_{11} = 1$

$13'3\, v_{12} = (28'7 - 18'7) v_{11} \longrightarrow v_{12} = \dfrac{10}{13'3} \longrightarrow v_{12} = 0'75$

$\boxed{\vec{v}_1 = \begin{bmatrix} 1 \\ 0'75 \end{bmatrix}}$

$\boxed{\lambda_2 = 0'9} \longrightarrow A\vec{v}_2 = \lambda_2 \vec{v}_2 = \begin{bmatrix} 18'7 & 13'3 \\ 13'3 & 10'9 \end{bmatrix} \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix} = 0'9 \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix} \longrightarrow$

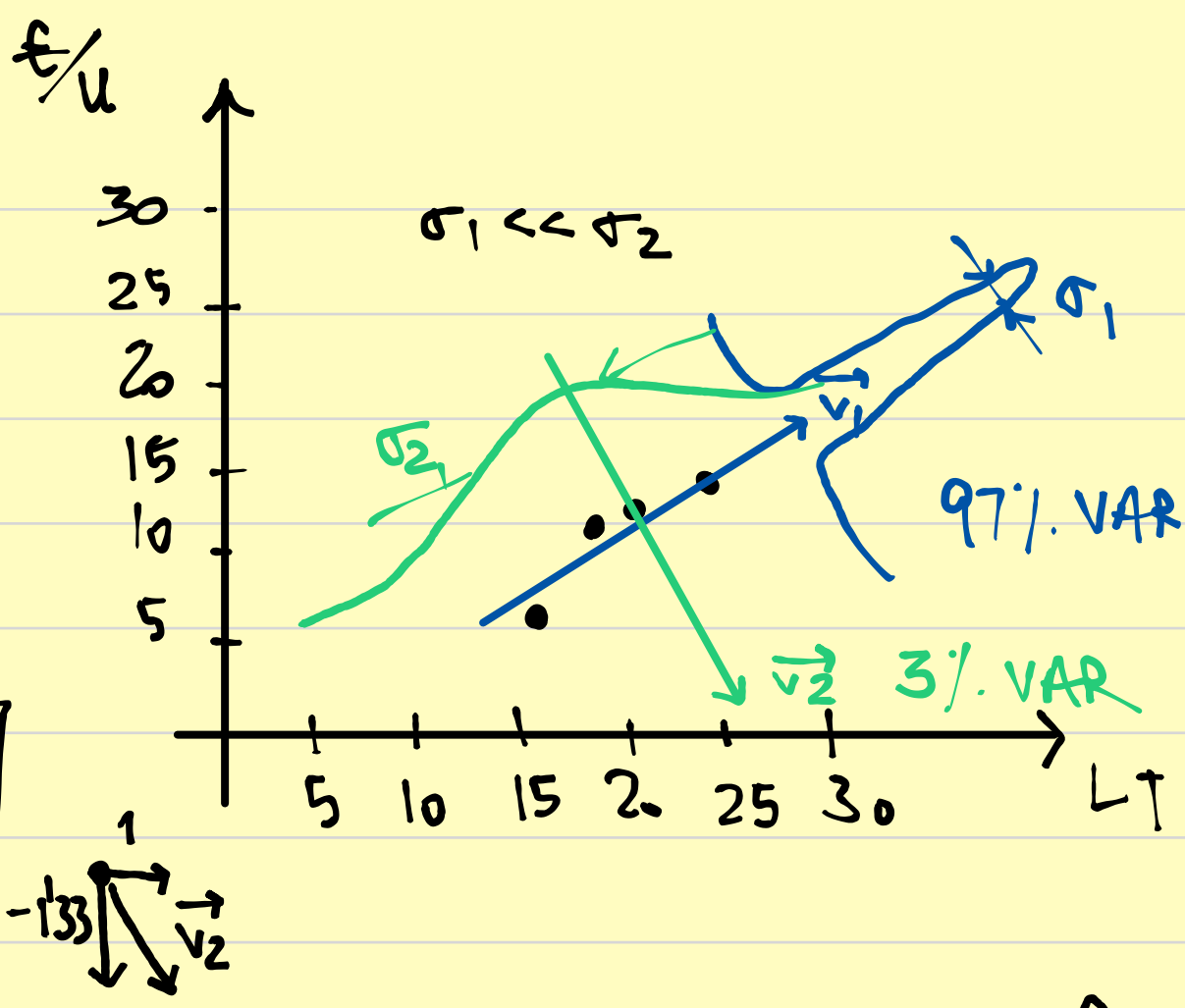
$18'7\, v_{21} + 13'3 \cdot v_{22} = 0'9\, v_{21}$

$13'3\, v_{21} + 10'9 \cdot v_{22} = 0'9\, v_{22}$ $\Rightarrow$ $v_{21} = 1$

$13'3\, v_{22} = (0'9 - 18'7) v_{21} \longrightarrow v_{22} = \dfrac{0'9 - 18'7}{13'3} = -1'33$

$\boxed{\vec{v}_2 = \begin{bmatrix} 1 \\ -1'33 \end{bmatrix}}$

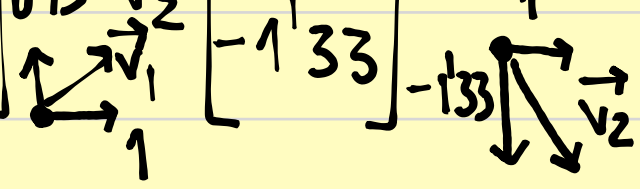
Graphical Representation:



LT: {11, 19, 21, 27}

€/u: {7, 11, 12, 15}

$\vec{v}_1 \begin{bmatrix} 1 \\ 1 \\ 0{,}75 \end{bmatrix}$ $0{,}75$ $\vec{v}_2 \begin{bmatrix} 1 \\ -1{,}33 \end{bmatrix}$ $-1{,}33$



Example: A company has a 3 dimensional Mgmt System (Lead Time, Cost, Quality). Please give the PCA and a management interpretation.

LT [Days]	{7, 11, 9, 13, 8}	≡ X
COST [€/Unit]	{317, 350, 327, 365, 330}	≡ Y
QUALITY [ppm]	{3718, 3970, 3780, 3790, 3810}	≡ Z

$$\text{VAR}: \frac{\sum (x_i - \bar{x})^2}{n-1} \qquad \text{COV}: \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

The first step to take is to be able to compare the KPIs. The fastest way to get the KPIs comparable is to normalize the series. Get them all between [0 & 1].

$$x_i^* = \frac{x_i - x_{min}}{x_{Mdx} - x_{min}} \quad [\text{NORMALIZATION}]$$

$LT^* . x^* . \left\{ \frac{7-7}{13-7}, \frac{11-7}{13-7}, \frac{9-7}{13-7}, \frac{13-7}{13-7}, \frac{8-7}{15-7} \right\} = \{0, 0'61, 0'33, 1, 0'17\}$

$Cost^* . Y^* . \left\{ \frac{317-317}{365-317}, \frac{350-317}{365-317}, \frac{327-317}{365-317}, \frac{365-317}{365-317}, \frac{330-317}{365-317} \right\} = \{0, 0'69, 0'21, 1, 0'27\}$

$Q^* . Z^* . \{0, 1, 0'24, 0'28, 0'36\}$

• • •

Cov. MAT.

$$A^* = \begin{bmatrix} 0'16 & 0'14 & 0'067 \\ 0'14 & 0'16 & 0'067 \\ 0'067 & 0'067 & 0'14 \end{bmatrix} \begin{matrix} x^* \\ y^* \\ z^* \end{matrix}$$

$x^* \quad y^* \quad z^*$

We approximate A^*, and make $0'067 \rightarrow 0$.

$$\tilde{A} = \begin{bmatrix} 0'16 & 0'14 & 0 \\ 0'14 & 0'16 & 0 \\ 0 & 0 & 0'14 \end{bmatrix} \rightarrow \det[\tilde{A} - \lambda I] = 0 \rightarrow$$

$$\rightarrow \det \begin{bmatrix} 0'16-\lambda & 0'14 & 0 \\ 0'14 & 0'16-\lambda & 0 \\ 0 & 0 & 0'14-\lambda \end{bmatrix} = 0 \rightarrow$$

$$\det \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix} = aei + dhc + bfg - ceg - bdi - afh$$

$$(0'14-\lambda) \cdot \left[(0'16-\lambda)^2 - 0'14^2 \right] = 0$$

$$\rightarrow \boxed{\lambda_2 = 0'14}$$

$$(0'16-\lambda)^2 - 0'14^2 = 0 \rightarrow \lambda^2 - 0'32\lambda + 0'006 = 0 \rightarrow$$

$$\lambda = \frac{0'32 \pm \sqrt{0'078}}{2} = \begin{cases} \lambda_1 = 0'3 \\ \lambda_3 = 0'02 \end{cases}$$

$$\text{VAR } \vec{v}_1 \,\% = \frac{0'3}{0'3+0'14+0'02} = 65'2\%$$

$$\text{VAR } \vec{v}_2 \,\% = \frac{0'14}{0'3+0'14+0'02} = 30'4\%$$

$$95'6\%$$

$$\text{VAR } \vec{v}_3 \,\% = \frac{0'02}{0'3+0'14+0'02} = 4'4\%$$

- $\lambda_1 = 0'3 \rightarrow \tilde{A}\vec{v}_1 = \lambda_1 \vec{v}_1 \rightarrow \begin{bmatrix} 0'16 & 0'14 & 0 \\ 0'14 & 0'16 & 0 \\ 0 & 0 & 0'14 \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{12} \\ v_{13} \end{bmatrix} = 0'3 \begin{bmatrix} v_{11} \\ v_{12} \\ v_{13} \end{bmatrix} \rightarrow$

$$0'16 \cdot v_{11} + 0'14 \cdot v_{12} = 0'3 v_{11} \rightarrow v_{11} = 1 \rightarrow v_{11} = v_{12}$$

$$0'14 v_{13} = 0'3 v_{13} \rightarrow v_{13} = 0$$

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

- $\lambda_2 = 0'14 \rightarrow \tilde{A}\vec{v}_2 = \lambda_2 \vec{v}_2 \rightarrow \begin{bmatrix} 0'16 & 0'14 & 0 \\ 0'14 & 0'16 & 0 \\ 0 & 0 & 0'14 \end{bmatrix} \begin{bmatrix} v_{21} \\ v_{22} \\ v_{23} \end{bmatrix} = 0'14 \begin{bmatrix} v_{21} \\ v_{22} \\ v_{23} \end{bmatrix}$

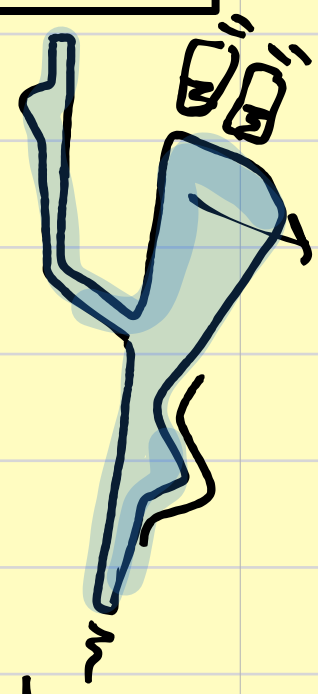
$$0'14 v_{23} = 0'14 v_{23} \rightarrow v_{23} = 1$$

$$0'16 v_{21} + 0'14 v_{22} = 0'14 v_{21} \rightarrow v_{21} = 1 \rightarrow v_{22} = -1$$

- $\lambda_3 = 0'02 \longrightarrow \ldots \longrightarrow \vec{v}_3 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$

$\vec{v}_2 = \begin{bmatrix} 1 \\ -7 \\ 1 \end{bmatrix}$

Villalba-Diez, J., Ordieres-Meré, J., Chudzick, H., Lopez-Rojo, P. (2015). NEMAWASHI: Attaining Value Stream alignment within Complex Organizational Networks. Procedia CIRP, 37, 134--139. https://doi.org/10.1016/j.procir.2015.08.021



NEMAWASHI. Prepare the ground.

n KPIs VISUALIZATION and its dynamics.

n KPIs ⟶ PCA ⟶ 3 PCs ⟶ Nemawashi

Example. A KPI System is 3 dimensional and 3 time series of data. Please implement NEMAWASHI to visualize and interpret organizational dynamics.

Quality [ppm]	{3300, 2700, 1800, 1500, 300}
Delivery Rate [%]	{91, 93, 89, 92, 95}
Cost [€/unit]	{17, 18, 16, 15, 16}

Step 1. Normalize each KPI btw [0&1]

$$q^* \left\{\frac{3300-1300}{3300-1300} = 1, 0'7, 0'25, 0'1, 0\right\} \in [0,1]$$

$$DR^* \left\{\frac{91-89}{95-89} = 0'33, 0'66, 0, 0'5, 1\right\} \in [0,1]$$

$$c^* \left\{\frac{17-15}{18-15} = 0'66, 1, 0'33, 0, 0'33\right\} \in [0,1]$$

Step 2. Normalize the temporal axis so that all the values add up to 1.

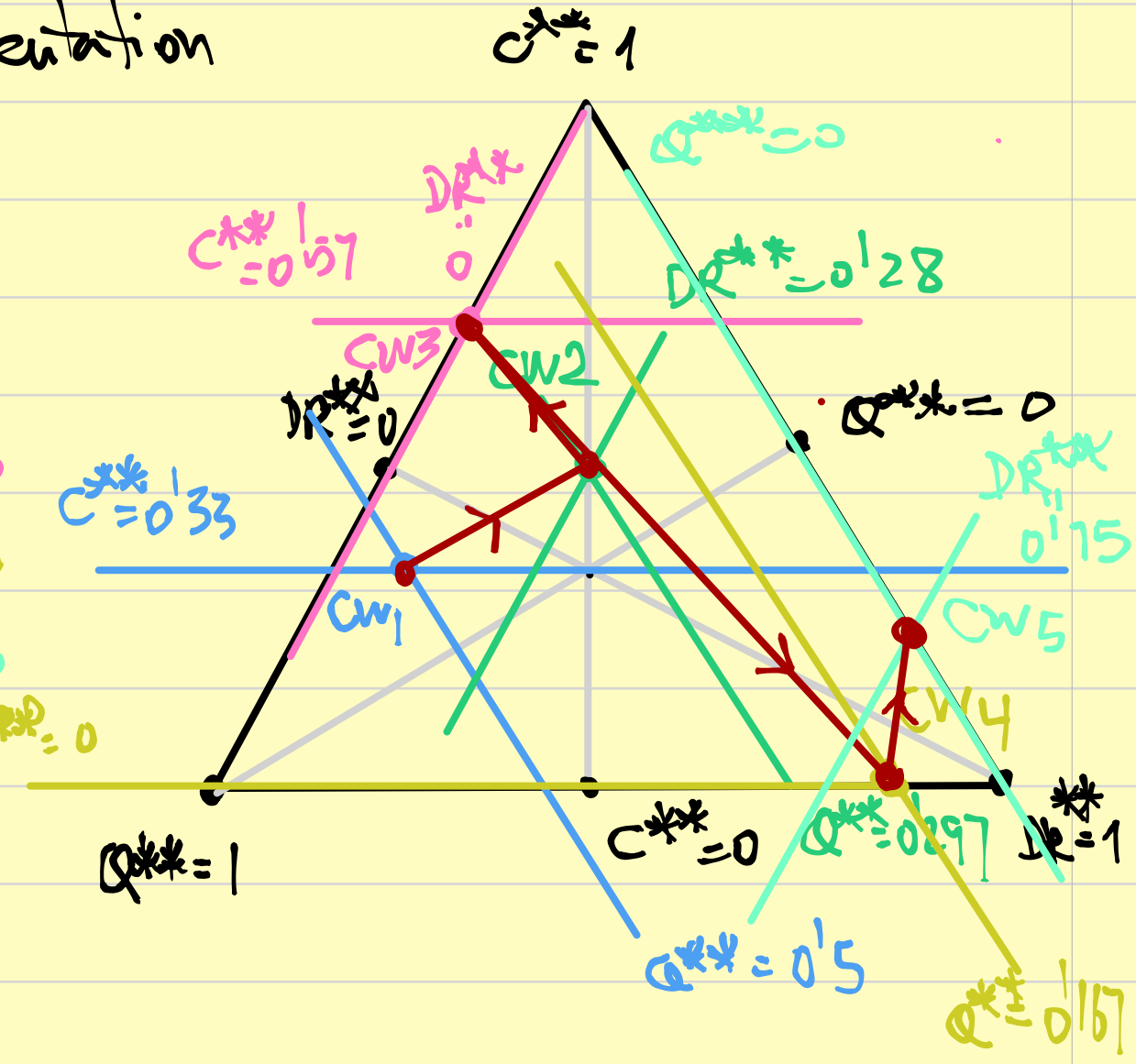
$$Q^{**}\left\{\frac{1}{1+0'33+0'66}=0'5\,;\ \frac{0'7}{0'7+0'66+1}=0'297\,;\ 0'43\,;\ 0'167\,;\ 0\right\}$$

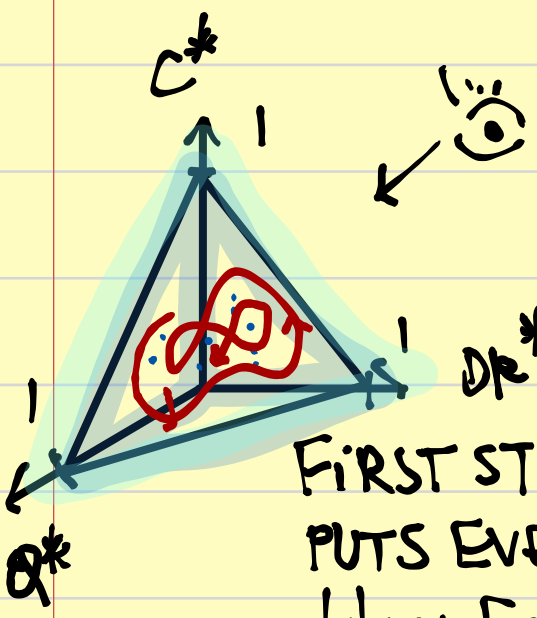
$$DR^{**}\left\{\frac{0'33}{1+0'33+0'66}=0'165\,;\ \frac{0'66}{0'7+0'66+1}=0'28\,;\ 0\,;\ 0'833\,;\ 0'75\right\}$$

$$C^{**}\left\{\frac{0'66}{1+0'33+0'66}=0'33\,;\ \frac{1}{0'7+0'66+1}=0'42\,;\ 0'57\,;\ 0\,;\ 0'25\right\}$$

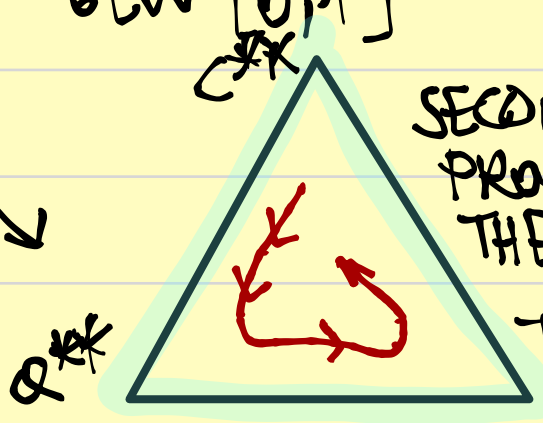
Step3. Graphical representation

	Q^{**}	DR^{**}	C^{**}	
CW_1	0'5	0'165	0'33	●
CW_2	0'297	0'28	0'42	●
CW_3	0'43	0	0'57	●
CW_4	0'167	0'833	0	●
CW_5	0	0'75	0'25	●





FIRST STEP PUTS EVERYPOINT btw [0,1]



SECOND STEP PROJECTS ALL THE POINTS ON THE SIMPLEX.

The condition for this projection is that the coordinates add up to 1.

MANAGEMENT INTERPRETATION OF THE DYNAMICS

The system is in Alignment on a timestamp t_j if the distance between $|t_{j-2}, t_{j-1}|$ is bigger than the distance between $|t_{j-1}, t_j|$.

$|CW_1 - CW_2| > |CW_2 - CW_3|$ → Distance decrease, so we have Alignment in CW3

$|CW_2 - CW_3| < |CW_3 - CW_4|$ → Distance increases, so we DO NOT have Alignment in CW

$|CW_3 - CW_4| > |CW_4 - CW_5|$ → Distance decrease, so we have Alignment in CW5.



within the blue circle, the organization is under control (in Alignment). The radius of this circle can be defined and is a mgmt decission.